

Miguel Aréjula Aísa

arejula10@gmail.com | +34 601 49 10 89 | LinkedIn | GitHub | are-dev.es

Profile

Software Engineering Master's student at SDU and former intern at **MongoDB** (Industry Solutions). Experienced in full-stack development, prototyping, and building practical **AI agents, tool workflows, and LLM integrations** using TypeScript, Next.js, and Python.

Experience

Industry Solutions Intern, MongoDB

July 2026 - September 2026

- Built an offline-first wallet demo for Financial Services: an on-device ObjectBox store synced bidirectionally with MongoDB Atlas, payments queued offline and replayed on reconnect, semantic transaction search on Voyage AI embeddings, and a LangGraph assistant on a local Ollama model.
- Maintained and extended the Next.js demo-deployment tool used across the Industry Solutions team, shipping bug fixes and new features to production.
- Wrote a Claude Code skill that drafts a full MongoDB Solutions Library article from a solution repository, encoding the team's template, taxonomy, and style rules so every write-up starts consistent instead of from a blank page.
- Shared the solution externally through a blog post, a MongoDB Solutions Library entry, and a YouTube video.

Teaching Assistant, SDU

February 2026 - May 2026

- Support students in understanding core software architecture concepts by assisting with lab sessions, assignments, and problem-solving, while clearly explaining technical concepts and collaborating with professors.

Software Engineer, University of Zaragoza

June 2024 – August 2025

- Researcher in project TED2021-130449B-I00 at the University of Zaragoza, where I led development of a custom web application for the Traumatology Department (Hospital Clínico Lozano Blesa), from requirements gathering with medical staff to architecture design and final delivery.
- Developed a full-stack system using React and PostgreSQL, with RESTful APIs in Express and FastAPI, streamlining operations and expected to support 50 patients per day, enhancing efficiency and quality of care.
- Integrated and processed structured clinical data (CSV-based surgical records) into a relational model, enabling generation and validation of clinical pathways through Petri nets and formal methods.

Projects

Leafy Wallet

github.com/leafy-wallet

- Offline-first personal wallet demo for Financial Services, where balances, activity, contacts, and the AI assistant all read from an on-device store and sync bidirectionally to MongoDB Atlas once the connection returns.
- Implemented offline payment queuing with replay on reconnect, multi-document ACID transactions to keep transfers consistent, and reconciliation against an external payment platform (PSP) through an OAuth PKCE SSO flow.
- Built semantic transaction search on Voyage AI embeddings, combining Atlas '\$rankFusion' hybrid search online with an on-device HNSW vector index offline, plus rolling-window fraud detection using '\$setWindowFields'.
- Developed Leafy, a LangGraph assistant running on a local Ollama model that routes questions to MongoDB MCP tools online and to the on-device store offline. **Tools Used:** Next.js, TypeScript, FastAPI, Python, MongoDB Atlas, ObjectBox, LangGraph, Ollama, Docker, Figma.

AgroNet

github.com/AgroNet

- Designed and developed a full-stack collaborative web platform for farmers, integrating real-time market prices, personalized weather alerts, and interactive geospatial data visualization across Spain.
- Led the front-end team, developed the web application, and deployed it on Vercel.
- Managed communication with the backend hosted on AWS, ensuring reliable API integration and real-time data synchronization. **Tools Used:** Astro, Node.js, TypeScript, Figma, MongoDB, AWS, Vercel.

ByronHUB

github.com/ByronHUB

- Developed ByronHub, a space booking application for the Ada Byron college building, implementing a four-tier hexagonal architecture with Domain-Driven Design principles.
- Set up containerized deployment using Docker Compose for local development and deployed the system to AWS and Railway with continuous deployment from GitHub.
- Designed and executed a testing strategy combining Postman, Jest unit/integration tests, and GitHub Actions for CI/CD pipelines, achieving automated validation and >80% coverage. Technologies: Node.js, TypeScript, React, PostgreSQL, RabbitMQ, Docker, AWS, Railway, Jest, GitHub Actions, Figma.

Skills & Tools

Languages & Core: TypeScript, JavaScript, Python, Go, C/C++, Java, SQL, HTML5, CSS3.
UI/UX & Frontend: React, Next.js, Astro, Tailwind CSS, Figma.
AI & Agentic Workflows: AI Agents (LangGraph, MCP tools), Tool Calling, Claude Code Skills, LLM Integration (Ollama, Claude API), Vector Search.
Backend, DB & Cloud: FastAPI, Node.js, Express, MongoDB Atlas, PostgreSQL, Docker, AWS, Vercel, REST APIs, Git.

Education

Master in Software Engineering , University of Southern Denmark	Sept 2025 – June 2027
Bachelor in Software Engineering , University of Zaragoza	Sept 2021 – June 2025

- Achieved high honors in Information Systems II and Verification and Validation, demonstrating exceptional proficiency and understanding in these subjects.